

Start at 9:32 am :)

ECON1050 Tutorial 4

Calculus 1: Limits and Differentiation

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Today's Plan

- ▶ Recap: key ideas from Lecture 4
- ▶ Tutorial questions: Q1–Q6

Recap: Limits

Core idea

A limit describes the value that $f(x)$ approaches when x approaches a point a .

Notation

$\downarrow \} \rightarrow \text{Constant; Convergent}$
 $\downarrow \} \rightarrow \pm\infty; \text{divergent}$

$\downarrow \} \rightarrow \text{Constant}$
 $\downarrow \} \rightarrow \pm\infty$

$$\lim_{x \rightarrow a} f(x).$$

$$\lim_{x \rightarrow a} f(x)$$

The limit depends on values of $f(x)$ close to a , not necessarily on the value of $f(a)$ itself.

Two-sided requirement

When computing a limit, values of x on both sides of a must be considered.

Recap: Univariate Differentiation

one-variable

Meaning

Differentiation measures the rate of change of one variable in response to a change in another variable. The derivative gives the slope of the tangent line to a function at a point.

Notation

Common notations include

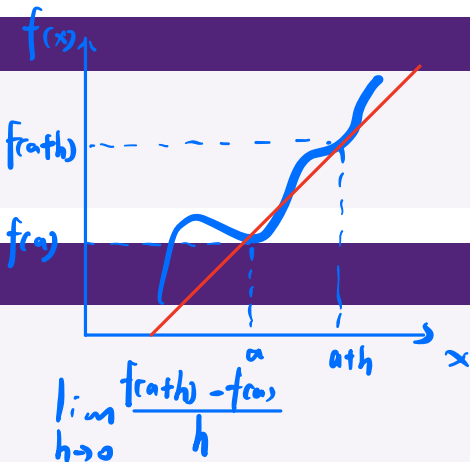
$$f'(x), \quad y', \quad \frac{dy}{dx}, \quad \frac{df(x)}{dx}.$$

$f'(x)$

Newton quotient

For a function $f(x)$, the derivative at a is

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}.$$



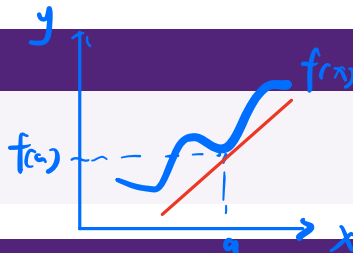
Recap: Tangent Line, Differentiability and Monotonicity

Tangent line

The tangent line to f at $x = a$ is

$$y - f(a) = f'(a)(x - a).$$

$$y - f(a) = f'(a)(x - a)$$



Differentiability

To be differentiable, a function must be continuous. Continuity is necessary, but not sufficient.

$$f \text{ is differentiable} \Rightarrow f \text{ is cts}$$

Using the first derivative on an interval I

$$\text{Increasing} \iff f'(x) \geq 0, \quad x \in I,$$

$$\text{Strictly increasing} \iff f'(x) > 0, \quad x \in I,$$

$$\text{Decreasing} \iff f'(x) \leq 0, \quad x \in I,$$

$$\text{Strictly decreasing} \iff f'(x) < 0, \quad x \in I.$$

Recap: Rates of Change

Instantaneous rate of change

The derivative gives the instantaneous rate of change:

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}.$$

Relative rate of change

The relative rate of change of f at a is

$$\frac{f'(a)}{f(a)}.$$

$$\frac{f'(a)}{f(a)}$$

Recap: Basic Differentiation Rules

Constant and power rules

$$\frac{d}{dx}c = 0, \quad \frac{d}{dx}x^a = ax^{a-1}.$$

Constants and sums

$$\frac{d}{dx}[cf(x)] = cf'(x), \quad \frac{d}{dx}[f(x) \pm g(x)] = f'(x) \pm g'(x).$$

Recap: Product, Quotient Rules, and Chain Rule

Product rule

If $F(x) = f(x)g(x)$, then

$$F(x) = f(x) \cdot g(x)$$

$$F'(x) = f'(x)g(x) + f(x)g'(x).$$

$$F'(x) = f' \cdot g + f \cdot g'$$

Quotient rule

If $F(x) = \frac{f(x)}{g(x)}$ and $g(x) \neq 0$, then

$$F(x) = \frac{f(x)}{g(x)}$$

$$F'(x) = \frac{f' \cdot g - g' \cdot f}{[g(x)]^2}.$$

Chain rule

Let $y = F(x) = f(g(x))$. Then

$$F(x) = f(g(x))$$

$$F'(x) = f'(g(x))g'(x).$$

Recap: Exponential and Logarithmic Derivatives

Exponential functions

$$\frac{d}{dx} e^x = e^x, \quad \frac{d}{dx} e^{g(x)} = g'(x) e^{g(x)}.$$

For base $a > 0$,

$$\frac{d}{dx} a^x = (\ln a) a^x.$$

Logarithmic functions

$$\frac{d}{dx} \ln x = \frac{1}{x}, \quad \frac{d}{dx} \ln(g(x)) = \frac{g'(x)}{g(x)}.$$

Recap: Higher Order Derivatives and Shape

Second derivative

If $f'(x)$ is differentiable, then its derivative is called the second derivative:

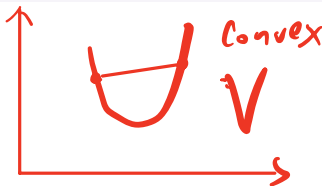
$$f''(x) = \frac{d^2 f(x)}{dx^2}.$$

Concavity and convexity

On an interval I ,

$$f''(x) \geq 0 \Rightarrow f \text{ is convex on } I,$$

$$f''(x) \leq 0 \Rightarrow f \text{ is concave on } I.$$



Tutorial Q1 & Q2

5 mins

Question 1

Compute the first derivative, i.e. $\frac{dy}{dx}$, for the following functions:

$$(a) \quad y = \frac{1}{3}x^9, \quad (b) \quad y = 3x^7 + 8, \quad (c) \quad y = 5x^3 + 3x - 1.$$

Question 2

Compute the first derivative for the following functions using the product rule or quotient rule:

$$(a) \quad y = (x^3 - 1)(x^2 + 2)$$

$$(b) \quad g(t) = \frac{2t + 1}{t^2 + 3}$$

$$(c) \quad y = 5x^2 + \frac{2x}{x^2 + 3}$$

$$(d) \quad y = (3x - 1)(2x^2 + 3)$$

$$Q1. (c) \ y = \underline{5x^3} + \underline{3x - 1}$$

$$= 5 \cdot 3 \cdot x^2 + 3$$

$$= 15x^2 + 3$$

$$Q2. (b) \ g(t) = \frac{2t+1}{t^2+3}$$

$$g(t) = \frac{f(t)}{h(t)} \Rightarrow g'(t) = \frac{f' \cdot h - f \cdot h'}{h^2}$$

$$f(t) = 2t+1 \Rightarrow f'(t) = 2$$

$$h(t) = t^2+3 \Rightarrow h'(t) = 2t$$

$$g'(t) = \frac{2 \cdot (t^2+3) - (2t+1) \cdot 2t}{(t^2+3)^2} = \frac{2t^2+6-4t^2-2t}{(t^2+3)^2} = \frac{-2t^2-2t+6}{(t^2+3)^2}$$

$$(d) \ y = (3x-1)(2x^2+3)$$

$$f(x) = g(x) \cdot h(x) \Rightarrow f' = g' \cdot h + g \cdot h'$$

$$g(x) = 3x-1 \Rightarrow g'(x) = 3$$

$$h(x) = 2x^2+3 \Rightarrow h'(x) = 4x$$

$$y' = 3 \cdot (2x^2+3) + (3x-1)4x = 6x^2+9 + 12x^2-4x \\ = 18x^2-4x+9$$

Tutorial Q1 & Q2: Knowledge Used

Knowledge used

- ▶ Constant rule: $\frac{d}{dx}c = 0$.
- ▶ Power rule: $\frac{d}{dx}x^a = ax^{a-1}$.
- ▶ Sum rule: differentiate each term separately.

Knowledge used

- ▶ Product rule:

$$(fg)' = f'g + fg'.$$

- ▶ Quotient rule:

$$\left(\frac{f}{g}\right)' = \frac{f'g - g'f}{g^2}, \quad g \neq 0.$$

- ▶ Sum rule: combine derivative rules term by term.

Tutorial Q1: Solution

Solution

$$(a) \quad y = \frac{1}{3}x^9 \quad \Rightarrow \quad y' = \frac{1}{3}(9)x^{9-1} = 3x^8.$$

$$(b) \quad y = 3x^7 + 8 \quad \Rightarrow \quad y' = 3(7)x^{7-1} + 0 = 21x^6.$$

$$(c) \quad y = 5x^3 + 3x - 1 \quad \Rightarrow \quad y' = 5(3)x^{3-1} + 3(1)x^{1-1} - 0 = 15x^2 + 3.$$

Tutorial Q2: Solution I

Part (a)

Let $f(x) = x^3 - 1$, $g(x) = x^2 + 2$. Then

$$\begin{aligned}y' &= f'(x)g(x) + f(x)g'(x) = 3x^2(x^2 + 2) + (x^3 - 1)(2x). \\&= 3x^4 + 6x^2 + 2x^4 - 2x = 5x^4 + 6x^2 - 2x.\end{aligned}$$

Part (b)

Let $f(t) = 2t + 1$, $h(t) = t^2 + 3$. Then $f'(t) = 2$ and $h'(t) = 2t$. Using the quotient rule,

$$\begin{aligned}g'(t) &= \frac{f'(t)h(t) - f(t)h'(t)}{[h(t)]^2} = \frac{2(t^2 + 3) - (2t + 1)(2t)}{(t^2 + 3)^2}. \\&= \frac{2t^2 + 6 - 4t^2 - 2t}{(t^2 + 3)^2} = \frac{-2t^2 - 2t + 6}{(t^2 + 3)^2}.\end{aligned}$$

Tutorial Q2: Solution II

Part (c)

$$\begin{aligned}y &= 5x^2 + \frac{2x}{x^2 + 3}. \\y' &= 10x + \frac{2(x^2 + 3) - 2x(2x)}{(x^2 + 3)^2}. \\&= 10x + \frac{2x^2 + 6 - 4x^2}{(x^2 + 3)^2} = 10x + \frac{6 - 2x^2}{(x^2 + 3)^2}.\end{aligned}$$

Part (d)

$$\begin{aligned}y' &= 3(2x^2 + 3) + (3x - 1)(4x) = 6x^2 + 9 + 12x^2 - 4x. \\y' &= 18x^2 - 4x + 9.\end{aligned}$$

Tutorial Q3

Question 3

5 mins

Compute the first derivative for the following functions using the chain rule:

(a) $y = 10(5 - x^2)^2$, let $u = 5 - x^2$,

(b) $y = (3x^2 + 6x + 6)^{1/2}$,

(c) $y = \sqrt{\frac{1}{x} - 1}$.

$$f(x) = h(g(x))$$

$$f'(x) = g'(x) \cdot h'(g(x)) \Rightarrow \begin{cases} g(x), h(x) \\ g'(x), h'(x) \\ g'(x) \cdot h'(g(x)) \end{cases}$$

$$(a) y = 10 \cdot (5 - x^2)^2 = 10 (g(x))^2$$

inner: $g(x) = 5 - x^2 \Rightarrow g'(x) = -2x$

outside: $h(x) = 10x^2 \Rightarrow h'(x) = 20x$

$$y' = g'(x) \cdot h'(g(x))$$

$$= (-2x) \cdot 20 \cdot (5 - x^2) = -200x + 40x^3$$

$$(c) y = \sqrt{\frac{1}{x} - 1} = \left(\frac{1}{x} - 1\right)^{\frac{1}{2}} = (g(x))^{\frac{1}{2}}$$

$$g(x) = \frac{1}{x} - 1 \Rightarrow g'(x) = -\frac{1}{x^2}$$

$$h(x) = x^{\frac{1}{2}} \Rightarrow h'(x) = \frac{1}{2} x^{-\frac{1}{2}} = \frac{1}{2} \cdot \frac{1}{\sqrt{x}}$$

$$y' = g'(x) \cdot h'(g(x))$$

$$= -\frac{1}{x^2} \cdot \frac{1}{2} \cdot \frac{1}{\sqrt{\frac{1}{x} - 1}}$$

$$= -\frac{1}{2x^2} \cdot \frac{1}{\sqrt{\frac{1}{x} - 1}}$$

Tutorial Q3: Knowledge Used

Knowledge used

- ▶ Chain rule for $y = f(g(x))$:

$$\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx}.$$

- ▶ Rewrite square roots as fractional powers.
- ▶ Apply the power rule to the outer function and differentiate the inner function.

Tutorial Q3: Solution I

Part (a)

Let

$$u(x) = 5 - x^2, \quad y = 10u^2.$$

Then

$$\frac{dy}{du} = 20u, \quad \frac{du}{dx} = -2x.$$

$$\frac{dy}{dx} = 20u(-2x) = 20(5 - x^2)(-2x).$$

$$\frac{dy}{dx} = 40x^3 - 200x.$$

Tutorial Q3: Solution II

Part (b)

Let

$$u = 3x^2 + 6x + 6, \quad y = u^{1/2}.$$

Then

$$\frac{dy}{du} = \frac{1}{2}u^{-1/2}, \quad \frac{du}{dx} = 6x + 6.$$

$$\frac{dy}{dx} = \frac{1}{2}(3x^2 + 6x + 6)^{-1/2}(6x + 6) = \frac{3x + 3}{(3x^2 + 6x + 6)^{1/2}}.$$

Tutorial Q3: Solution III

Part (c)

$$y = \sqrt{\frac{1}{x} - 1} = \left(\frac{1}{x} - 1\right)^{1/2}.$$

Let

$$u = \frac{1}{x} - 1, \quad \frac{du}{dx} = -x^{-2}, \quad y = u^{1/2}.$$

$$\frac{dy}{dx} = \frac{1}{2} u^{-1/2} (-x^{-2}) = -\frac{1}{2} \left(\frac{1}{x} - 1\right)^{-1/2} x^{-2}.$$

$$\frac{dy}{dx} = -\frac{1}{2\sqrt{x^3 - x^4}}.$$

Tutorial Q4

Question 4

Compute the second derivatives for the following functions:

5 mins + 30 seconds

(a) $z = 120t - \frac{1}{3}t^3$, (b) $y = \sqrt{x}$, (c) $g(t) = \frac{t^2}{t-1}$.

(a) $z = 120t - \frac{1}{3}t^3$
 $z' = 120 - \frac{1}{3} \cdot 3 \cdot t^2$
 $= 120 - t^2$
 $z'' = 0 - 2t$
 $= -2t$

(c) $g(t) = \frac{t^2}{t-1}$
 $g'(t) = \frac{2t \cdot (t-1) - t^2 \cdot 1}{(t-1)^2} = \frac{2t^2 - 2t - t^2}{(t-1)^2} = \frac{t^2 - 2t}{(t-1)^2}$
 $g''(t) = \frac{(t^2 - 2t)' \cdot (t-1)^2 - (t^2 - 2t) \cdot (t-1)'}{(t-1)^4}$
 $= \frac{(2t-2)(t-1)^2 - (t^2-2t) \cdot 2 \cdot (t-1)}{(t-1)^4}$
 $= \frac{2(t-1)^2 - 2(t^2-2t)}{(t-1)^3} \quad \text{Cancel } t-1$
 $= \frac{2t^2 - 4t + 2 - 2t^2 + 4t}{(t-1)^3} = \frac{2}{(t-1)^3}$

Tutorial Q4: Knowledge Used

Knowledge used

- ▶ First derivative: differentiate the original function once.
- ▶ Second derivative: differentiate the first derivative.
- ▶ Use the power rule and quotient rule where needed.

Tutorial Q4: Solution I

Part (a)

$$z = 120t - \frac{1}{3}t^3.$$

$$z' = 120 - \frac{1}{3}(3)t^2 = 120 - t^2.$$

$$z'' = -2t.$$

Part (b)

$$y = \sqrt{x} = x^{1/2}.$$

$$y' = \frac{1}{2}x^{-1/2}.$$

$$y'' = -\frac{1}{4}x^{-3/2} = -\frac{1}{4\sqrt{x^3}}.$$

Part (c)

$$g(t) = \frac{t^2}{t-1}, \quad t \neq 1.$$

$$g'(t) = \frac{2t(t-1) - t^2}{(t-1)^2} = \frac{t^2 - 2t}{(t-1)^2}.$$

$$\begin{aligned} g''(t) &= \frac{(2t-2)(t-1)^2 - 2(t-1)(t^2-2t)}{(t-1)^4} \\ &= \frac{(2t-2)(t-1) - 2(t^2-2t)}{(t-1)^3} = \frac{2}{(t-1)^3}. \end{aligned}$$

Question 5

- ① Find the intervals where the following function is strictly increasing: $\Leftrightarrow f'(x) > 0$

$$y = (\ln x)^2 - 4.$$

- ② For the function

$$f(x) = 2x^3 - 12x^2 + 5,$$

find the intervals where it is increasing and the intervals of concavity/convexity.

$$(\ln x)' = \frac{1}{x}$$

$$(1). y = (\ln x)^2 - 4$$

$$y' = \frac{1}{x} \cdot 2 \cdot \ln x = \frac{2 \ln x}{x} > 0$$



$$f'(x) \geq 0$$

sign diagram

	$(0, 1)$			$(1, \infty)$
$\ln x$	-	0	+	
x	+	+	+	
$\frac{2 \ln x}{x}$	-	0	+	

$$x \in (1, \infty)$$

$$f''(x) \leq 0$$

$$f''(x) \geq 0$$

$$(2). \quad f(x) = 2x^3 - 12x^2 + 5$$

$$f'(x) = 6x^2 - 24x$$

$$f''(x) = 12x - 24$$

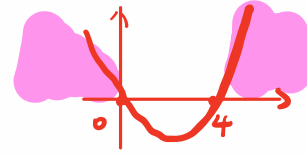
$$\text{Increasing} \Leftrightarrow f'(x) \geq 0 \Leftrightarrow 6x^2 - 24x \geq 0$$

$$\underline{x \in \mathbb{R} \setminus (0, 4)}$$

$$6x^2 - 24x = 0$$

$$x(x - 4) = 0$$

$$x = 0, 4$$



$$f''(x) = 12x - 24$$

$$\text{Concave: } f''(x) \leq 0 \Leftrightarrow 12x - 24 \leq 0 \Leftrightarrow \underline{x \leq 2}$$

$$\text{Convex: } f''(x) \geq 0 \Leftrightarrow 12x - 24 \geq 0 \Leftrightarrow \underline{x \geq 2}$$

Tutorial Q5: Knowledge Used

Knowledge used

- ▶ A function is strictly increasing where $f'(x) > 0$.
- ▶ A function is increasing where $f'(x) \geq 0$.
- ▶ A function is concave where $f''(x) \leq 0$ and convex where $f''(x) \geq 0$.
- ▶ Use sign diagrams to identify intervals.

Tutorial Q5: Solution I

Part 1

$$y = (\ln x)^2 - 4, \quad x > 0.$$

$$y' = 2(\ln x) \frac{1}{x} = \frac{2 \ln x}{x}.$$

Since $x > 0$, the sign of y' is determined by $\ln x$:

x	$(0, 1)$	1	$(1, \infty)$
$\ln x$	$-$	0	$+$
x	$+$	$+$	$+$
y'	$-$	0	$+$

Therefore,

y is strictly increasing for $x > 1$.

Tutorial Q5: Solution II

Part 2: Increasing intervals

$$f(x) = 2x^3 - 12x^2 + 5.$$

$$f'(x) = 6x^2 - 24x = 6x(x - 4).$$

x	$(-\infty, 0)$	0	$(0, 4)$	4	$(4, \infty)$
x	$-$	0	$+$	$+$	$+$
$x - 4$	$-$	$-$	$-$	0	$+$
$f'(x)$	$+$	0	$-$	0	$+$

Thus

$f(x)$ is increasing on $(-\infty, 0] \cup [4, \infty)$.

Tutorial Q5: Solution III

Part 2: Concavity and convexity

$$f'(x) = 12x - 24 = 12(x - 2).$$

x	$(-\infty, 2)$	2	$(2, \infty)$
$x - 2$	$-$	0	$+$
$f'(x)$	$-$	0	$+$

Since $f'(x) \leq 0$ for $x \leq 2$ and $f'(x) \geq 0$ for $x \geq 2$,

$f(x)$ is concave on $(-\infty, 2]$

and convex on $[2, \infty)$.

Tutorial Q6

Question 6

For the function

$$y = \sqrt{x} - x^2,$$

find the equation for the tangent to the graph at the point $x = 4$.

General form for tangent line: $y - f(a) = f'(a) \cdot (x - a)$ $a = 4$

$$y - f(4) = f'(4) (x - 4)$$

$$f(x) = \sqrt{x} - x^2 \Rightarrow f(4) = \sqrt{4} - 4^2 = 2 - 16 = -14$$

$$f'(x) = \frac{1}{2} \cdot x^{-\frac{1}{2}} - 2x = \frac{1}{2} \cdot \frac{1}{\sqrt{x}} - 2x \Rightarrow f'(4) = \frac{1}{2} \cdot \frac{1}{\sqrt{4}} - 2 \cdot 4 = \frac{1}{4} - 8 = -\frac{31}{4}$$

$$\Rightarrow \text{tangent line: } y + 14 = -\frac{31}{4} (x - 4)$$

Tutorial Q6: Knowledge Used

Knowledge used

► The tangent slope at $x = a$ is $f'(a)$.

► Tangent line formula:

$$y - f(a) = f'(a)(x - a).$$

► Use the power rule to differentiate $\sqrt{x} = x^{1/2}$.

Tutorial Q6: Solution

Solution

$$f(x) = x^{1/2} - x^2.$$

$$f'(x) = \frac{1}{2}x^{-1/2} - 2x.$$

At $x = 4$,

$$f'(4) = \frac{1}{2}(4)^{-1/2} - 2(4) = \frac{1}{2} \cdot \frac{1}{\sqrt{4}} - 8 = \frac{1}{4} - 8 = -\frac{31}{4}.$$

$$f(4) = \sqrt{4} - 4^2 = 2 - 16 = -14.$$

Using $y - f(4) = f'(4)(x - 4)$:

$$y + 14 = -\frac{31}{4}(x - 4).$$

Therefore,

$$y = -\frac{31}{4}x + 17.$$

Key Takeaways

- ▶ The derivative measures instantaneous rate of change and tangent slope.
- ▶ Product, quotient and chain rules are essential for differentiating composite expressions.
- ▶ The first derivative identifies increasing/decreasing intervals.
- ▶ The second derivative identifies concavity and convexity.